

Thesis Update

Evaluating Foundation Models for Financial Time-Series Forecasting

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November 5, 2025

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Outline

Research Questions

Models

Data

Experimental Design

Current Progress

Next Steps

Research Questions

Research Questions

RQ1: Forecast Horizon Length Effects

- How does forecast horizon length affect the relative performance of foundation models compared to traditional approaches?

RQ2: Temporal Frequency Effects

- How does temporal frequency of input data affect the relative performance of foundation models versus traditional approaches?

RQ3: Market Disruption Resilience

- How do foundation models compare to traditional approaches in maintaining accuracy during high-volatility market regimes?

Models

Models Under Evaluation

Model	Type	Description
Naive	Baseline	Simple persistence model using last observation
ARIMA	Statistical	Classical auto-regressive model with differencing
Chronos	Foundation	Transformer-based pretrained model (Amazon)
Sundial	Foundation	Zero-shot forecasting with pretraining
TimesFM	Foundation	Google's large-scale time-series foundation model

Data

Dataset

Stock: Microsoft (MSFT)

Time Frequencies:

- **Hourly (1h):** High-frequency intraday trading data
- **Daily (1d):** End-of-day closing prices

Test Period: 2023

Training Periods:

- Short-term: 25-250 days (linear and logarithmic spacing)
- Long-term: 1-10 years

Data Transformations

Two transformations tested:

- **Raw prices:** Original closing prices
- **Percentage change:** Relative change between consecutive observations

Percentage Change Formula:

$$\text{Relative Change} = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}}$$

Why Percentage Change?

- **Stationarity:** Percentage changes are more stationary than raw prices
- **Financial relevance:** Returns are more meaningful than absolute prices for trading decisions

Experimental Design

Experimental Design

Training Configurations:

- **Short-term:** 25-250 days
 - Linear spacing
 - Logarithmic spacing
- **Long-term:** 1-10 years

Forecast Horizons:

- Hourly: 1 hour
- Daily: 1, 5, 21, 63 days

Evaluation Metrics:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- MAPE (Mean Absolute % Error)
- MSE (Mean Squared Error)
- MDA (Mean Directional Accuracy)

Transformations:

- Raw prices
- Percentage change

Forecasting Approach

Rolling forecast updating context with actual observations

Current Progress

Experiments Completed

- **All models implemented and tested:**
 - Traditional: ARIMA, Naive baseline
 - Foundation models: Chronos, Sundial, TimesFM
- **Comprehensive experimental setup:**
 - Multiple training configurations (25-250 days, 1-10 years)
 - Multiple forecast horizons (1h, 1d, 5d, 21d, 63d)
 - Two temporal frequencies (hourly, daily)
 - Two data transformations (raw prices, percentage change)

Next Steps

RQ2: Temporal Frequency Effects

- Compare model performance across hourly, and daily frequencies
- Analyze how different temporal resolutions affect foundation vs traditional models

RQ3: Market Disruption Resilience

- Identify high-volatility market regimes (e.g., COVID-19 crash, earnings announcements)
- Evaluate model performance during market disruptions
- Compare stability of foundation models vs traditional approaches under stress

Next:

- Complete results analysis
- RQ3: High-volatility regime analysis
- Start writing experiments section